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Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE498 – Project Phase II (Technical Progress Report - Review 3)
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1 Project Objectives Completed

In this final phase of the Capstone project, we have fully designed, implemented, and verified the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**. The core objectives completed are:

- **Privacy-Preserving Collaborative Training:** Mastered and coded decentralized training using standard *FedAvg* and regularization-based *FedProx* in PyTorch across simulated hospital clients, ensuring raw patient data remains localized.
- **Personalized Drift Adaptation:** Built custom client-side personalization layers combined with a statistical **AD-CUSUM Residual Neural Trigger**. This unfreezes local classifier heads dynamically when institutional data drift is detected, reducing communication overhead.
- **Dual-Level Clinical Explainability:** Embedded a temporal self-attention mechanism to output patient-level attention weights over a 24-hour sequence window, pointing clinicians directly to critical timestamps.
- **Interactive CDSS Workstation:** Developed a production-quality Clinical Decision Support System (CDSS) dashboard using React, TypeScript, FastAPI, and a relational SQLite database for real-time risk predictions.

1.1 Core Technical Novelties

Our proposed FPDAF framework introduces three key scientific novelties that address the limitations of conventional healthcare federated setups:

1. **Sequential Attention Interpretability:** Unlike static black-box networks, we combine LSTM backbones with query-based temporal self-attention to output dynamic sequence saliency weights, allowing clinicians to visually track when a patient's risk profile escalates.
2. **AD-CUSUM-Triggered Drift Adaptation:** We replace the standard continuous personalized weight aggregation model with an online Cumulative Sum (AD-CUSUM) monitor that audits prediction loss residuals on the fly. Nodes adjust local head adaptation rates only when statistically significant concept drift occurs.

3. **Client-Side Selective Personalization (CSSP):** Upon AD-CUSUM drift detection, the framework triggers CSSP to freeze the shared temporal LSTM layers and fine-tune only the local classification head. This reduces model communication overhead by **38%** (saving network bandwidth) and cuts node adaptation time to under 6 minutes.

2 Clinical Profile of ICU Sepsis & Significance

To establish the clinical context and motivation for this project, we outline the disease pathology and the specific public health challenge it presents:

- **What Sepsis is:** Sepsis is a life-threatening medical emergency caused by the body's extreme, systemic immune response to an infection, which ends up damaging its own tissues, blood vessels, and vital organs.
- **Why it is Dangerous:** If not diagnosed immediately, it rapidly progresses to Septic Shock, leading to multi-organ failure, a sudden drop in blood pressure, and high mortality rates.
- **The Clinical Time-Window Challenge:** Sepsis is notoriously difficult to detect early because its initial symptoms resemble general infections. However, every single hour of delayed treatment increases a patient's risk of death by approximately **8%**.
- **Role of Our Project:** By continuously analyzing ICU vital signals, our FPDAF early-warning system acts as an early-warning radar system, alerting doctors hours before a patient clinically deteriorates, giving clinicians a vital head-start to administer antibiotics and fluids.

2.1 Sepsis Burden & Impact in India

Sepsis represents a critical public health crisis in India, necessitating real-time, automated early warning systems inside intensive care units:

- **National Mortality & Case Load:** Sepsis accounts for approximately **2.9 to 3.0 million deaths** annually in India from an estimated **11.0 to 11.3 million active cases** [?]. It contributes to nearly **30% (3 in 10)** of all deaths in the country.
- **Indian ICU Mortality Rates:** Severe sepsis patients admitted to Indian intensive care units (ICUs) face an extremely high mortality rate ranging from **34% to over 50%** [?]. This is heavily compounded by rising rates of antimicrobial resistance (AMR), which makes early diagnostic windows crucial.

3 Data Engineering & Imputation Pipeline

The system targets the **PhysioNet Challenge 2019 Sepsis dataset**. Sparsity and variable logging rates present major data engineering issues (e.g., heart rate logged hourly, while bilirubin is sampled daily). Our completed pipeline processes raw patient files ('.psv') as follows:

1. **Forward-Filling (Time-Series Carry):** Carries the last recorded measurement forward.
2. **Backward-Filling & Zero-Padding:** Imputes initial baseline conditions and fills remaining missing blocks with 0.0 to ensure structural uniformity.
3. **Demographics Extraction:** Extracts patient age and gender.
4. **Normalization**: Fits a leakage-free 'StandardScaler' strictly on the training partition and saves it to 'scaler.pkl' to standardize features before feeding them into PyTorch.

4 Neural Architecture & Drift Adaptation Formalism

FPDAF splits model weights into shared backbones (temporal feature extraction) and private local heads (classification). The mathematical formulation is:

$$\hat{y}_t^k = h_{\phi^k}\left(\sum_{i=1}^{t} \alpha_i \cdot g_\theta(X_i^k)\right)$$

where g_θ represents the shared global LSTM layers (weights θ), h_{ϕ^k} is the local classifier head (weights ϕ^k), and α_i are temporal self-attention coefficients computed as:

$$e_i = v_a^T \tanh(W_s s_i + b_a), \quad \alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{t} \exp(e_j)}$$

To handle temporal drift without constant full-parameter aggregation, our team developed a novel statistical drift detection algorithm: **Adaptive Divergence-Weighted CUSUM (AD-CUSUM)**. It monitors multi-modal validation error progression at communication round r:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left(\mathcal{L}_{val,r} + \beta \cdot D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$$

where $D_{\text{JSD}}^{(r)} = \text{JSD}(P^{(r)} \parallel P^{(r-1)})$ measures prediction entropy divergence across rounds, $w_{\text{grad}}^{(r)} = 1 + \tanh(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L}))$ scales updates by classifier gradient variance, and $\kappa_r = \mu_{loss} + \alpha\sigma_{loss}$ provides auto-calibrating dynamic slack. If $S_r > H$ (threshold $H = 3.0$), the framework declares concept drift, freezes the global backbone θ, and adapts the local personalization head ϕ^k independently (Client-Side Selective Personalization - CSSP).

Algorithm 1 Developed AD-CUSUM Drift Detection Algorithm (Project Innovation)

Require: Validation loss $\mathcal{L}_{\text{val},r}$, prediction probabilities $P^{(r)}$, baseline μ_0, threshold $H = 3.0$.
Ensure: Drift alert boolean drift_flag, updated running score S_r.

Compute Dynamic Slack: $\kappa_r \leftarrow \max(0.01, \mu_{\text{loss}} + \alpha \cdot \sigma_{\text{loss}} - \mu_0)$
if $P^{(r-1)}$ exists **then**
  $D_{\text{JSD}}^{(r)} \leftarrow \text{JSD}\left(P^{(r)} \parallel P^{(r-1)}\right)$
else
  $D_{\text{JSD}}^{(r)} \leftarrow 0.0$
end if
Compute Gradient Variance Weight: $w_{\text{grad}}^{(r)} \leftarrow 1.0 + \tanh\left(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L})\right)$
Calculate Recurrence Update: $S_r \leftarrow \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{\text{val},r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$
if $S_r > H$ **then**
  drift_flag $\leftarrow$ True, $S_r \leftarrow 0.0$ {Trigger CSSP local head fine-tuning}
else
  drift_flag $\leftarrow$ False
end if
return drift_flag, S_r

5 Five-Way Benchmarking & Results Analysis

We evaluated five different training setups on the preprocessed test set. The experimental metrics are summarized in Table 1:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (Proposed)	**86.51%**	**9.81%**	**65.33%**	**0.1706**	**0.8214**	**1.52 GB (CSSP saved 38%)**

Table 1: Decentralized Model Benchmark Results (5-Way Comparison)

Analysis: While conventional federated models suffer from performance drops under demographic heterogeneity, our proposed FPDAF framework achieves the highest overall decentralized performance, yielding an Accuracy of **86.51%**, F1-score of **0.1706**, and AUROC of

0.8214 (an improvement of **+2.25%** in AUROC and **+1.32%** in F1-score over Ditto Personalized). At the same time, FPDAF saves **38% in communication bandwidth** compared to Ditto (1.52 GB vs 2.48 GB) because local drift adaptation is head-only and occurs only when triggered.

6 Ablation Studies Analysis

We verified the components of FPDAF by stripping different modules. The results are detailed in Table 2:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o AD-CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	**86.51%**	**9.81%**	**65.33%**	**0.1706**	**0.8214**

Table 2: Ablation Study Results

7 Clinician CDSS Workspace Application Architecture

We built a production-quality Clinical Decision Support System (CDSS) workstation. The architecture is composed of:

1. **SQLite Relational DB (‘clinical_warehouse.db‘):** Houses tables for ‘patients‘ (demographics), ‘vitals‘ (24h metrics decoded using ‘scaler.pkl‘), ‘predictions‘ (dynamic PyTorch inferences), ‘attentions‘ (temporal coefficients), and ‘drift_metrics‘ (AD-CUSUM values).
2. **FastAPI REST Backend (‘main.py‘):** Provides API endpoints for patient records, vital timeline charts, and attention weights.
3. **React Vite Frontend:** Features custom views for **Doctors** (patient bed overview, risk predictor dials, and XAI explainers) and **Admins** (concept drift AD-CUSUM charts, federated aggregation loop, and ROC curve comparison overlays).

8 Summary of works completed in Phase II

All milestones on the project roadmap have been successfully met:

1. **Imputation Pipeline:** Imputed, standardized, and saved the test dataset.
2. **Baseline Implementation:** Coded and trained FedAvg, FedProx, and Ditto baselines.
3. **Proposed Model Training:** Trained the FPDAF model with local attention queries.
4. **DB & API Service:** Implemented a database-driven FastAPI backend.
5. **Web Workstation UI:** Completed a responsive clinician dashboard with dark-mode support.

Appendix: Data & Resource Access

Phase II Project Repository: `https://github.com/aksharsakhi/23CSE498-Project-Phase-II.git`

FastAPI Backend & DB Schema: `https://github.com/aksharsakhi/23CSE498-Project-Phase-II/tree/main/backend`

React Frontend Workspace: `https://github.com/aksharsakhi/23CSE498-Project-Phase-II/tree/main/dashboard`